Applied Artificial Intelligence and Machine Learning

Course Syllabus & Learning Outcomes

COURSE INFORMATION

Course Overview

Course Code: Applied AI and ML (AIML)

Total Duration: 12 Weeks

Total Hours: 120 Hours

This course offers a comprehensive introduction to AI and machine learning fundamentals, covering supervised learning, unsupervised learning, neural networks, and practical applications.

LEARNING GOALS

Course Learning Outcomes

1

Understand fundamental AI and ML concepts, algorithms, and their real-world applications

2

Build and train machine learning models using Python and popular libraries

3

Apply supervised and unsupervised learning techniques to solve practical problems

4

Evaluate model performance and implement best practices in ML development

COURSE SCHEDULE

Weekly Learning Modules

Week 1: Foundations & Python Basics

10 hours

Week 3: Python for Data Science

10 hours

Week 5: Machine Learning Algorithms

10 hours

Week 4: Evaluation Matrix and Ensemble Learning

10 hours

Week 5: Unsupervised Learning

10 hours

Week 6: Capstone Project & Real-world Applications

10 hours

WEEK 1

Introduction of AI and Fundamentals Of Python

Content Delivery Duration: 20 Hours

01

Day 1

Introduction to Artificial Intelligence and Machine Learning - Applications across domains (business, healthcare, automation, vision, language), Overview of the ML pipeline, types of AI (narrow, general), Key AI problems and techniques, Modern AI Toolkits (TensorFlow, PyTorch) | 2 Hours

02

Day 2

Python basics: Variables, data types, operators, Control flow in Python, Conditional statements, Loops, Variables, data types, operators and practice implementations | 2 Hours

03

Day 3

Python data structures: Lists, tuples, sets, dictionaries | 2 Hours

04

Day 4

Functions, Modules and packages, File handling | 2 Hours

05

Day 5

Introduction to NumPy, Arrays and operations, Mathematical operations with NumPy | 2 Hours

Total: 10 Hours

WEEK 2

Python for Data Science

Content Delivery Duration: 10 Hours

01

Day 6

Introduction to Pandas, DataFrames and Series, Practical implementation | 2 Hours

02

Day 7

EDA: Data visualization using Matplotlib & Seaborn | 2 Hours

03

Day 8

Types of data, Data pre-processing, data leakage concept, handling missing values, outliers handling, handling categorical data, scaling and normalization, class imbalance handling | 2 Hours

04

Day 9

Feature Engineering: variance threshold Feature Selection, correlation-based removal Feature Selection, forward selection, backward elimination, tree-based feature importance | 2 Hours

05

Day 10

Feature Extraction, creating new features, aggregation features, Dimensionality Reduction: Curse of dimensionality, Principal Component Analysis, explained variance ratio, Linear Discriminant Analysis | 2 Hours

Total: 10 Hours

WEEK 3

Machine Learning Algorithms

Content Delivery Duration: 10 Hours

01

Day 11

Introduction to Machine Learning algorithms, Linear Regression with practical implementation | 2 Hours

02

Day 12

Logistic Regression: Classification problems (Maths behind Logistic algorithm and practical implementation) | 2 Hours

03

Day 13

Types of Linear Regression and Regularization, Different types of Gradient Descent | 2 Hours

04

Day 14

K-Nearest Neighbors (KNN), Decision Tree Algorithm and Random Forest | 2 Hours

05

Day 15

CA 1 (Continuous Assessment 1) | 2 Hours

Total: 10 Hours

WEEK 4

Evaluation Matrix and Ensemble Learning

Content Delivery Duration: 10 Hours

01

Day 16

Support Vector Machine and Model evaluation: Confusion Matrix, Accuracy, Precision, Recall, F1 Score, ROC Curve, Precision-Recall Curve, Cross Validation | 2 Hours

02

Day 17

Ensemble Learning: Introduction to ensemble learning, Bagging & Boosting Ensembles | 2 Hours

03

Day 18

Cross Validation and Hyperparameter tuning | 2 Hours

04

Day 19

Project-1 | 2 Hours

05

Day 20

Project-2 | 2 Hours

Total: 10 Hours

WEEK 5

Unsupervised Learning

Content Delivery Duration: 10 Hours

01

Day 21

Unsupervised Learning: Role of unsupervised learning, Differences between supervised and unsupervised learning, Euclidean, Manhattan, Cosine distances, Choosing appropriate distance metrics | 2 Hours

02

Day 22

K-Means Clustering, Elbow method, K-Medoids, Hierarchical Clustering | 2 Hours

03

Day 23

Introduction to Deep Learning, Neural Networks, Neurons and layers | 2 Hours

04

Day 24

Activation functions, ReLU, Sigmoid, Tanh, Loss functions and optimizers, Gradient Descent, SGD, Adam | 2 Hours

05

Day 25

ANN and Practical Implementation with different Deep learning tools | 2 Hours

Total: 10 Hours

WEEK 6

CNN and RNN

Content Delivery Duration: 10 Hours

01

Day 26

CNN: Filters (kernels), Stride and padding, Activation functions (ReLU), Pooling layers (Max Pooling, Average Pooling), Flatten layer | 2 Hours

02

Day 27

RNN: types, Sequential data and time-series data, Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU) | 2 Hours

03

Day 28

Project-3 | 2 Hours

04

Day 29

CA 2 (Continuous Assessment 2) | 2 Hours

05

Day 30

Project-4 | 2 Hours

Total: 10 Hours

EVALUATION

Assessment Methods & Grading Breakdown

20%
Participation & Quizzes

40%
Assignments & Projects

15%
Midterm Exam

25%
Final Capstone Project

All assessments are designed to reinforce learning outcomes and practical skills.

RESOURCES

Learning Materials & Tools

Required Textbooks

"Hands-On Machine Learning" and "Deep Learning" by Goodfellow

Software & Tools

Python 3.9+, Jupyter Notebook, scikit-learn, TensorFlow, Pandas

Online Platforms

Kaggle, GitHub, Google Colab for hands-on practice

Support Resources

Office hours, discussion forums, peer study groups

COURSE POLICIES

Expectations & Academic Integrity

Student Expectations

- Attend all live sessions and complete weekly assignments
- Engage actively in discussions and peer learning
- Maintain academic integrity in all work

Course Policies

- Late submissions accepted with 10% penalty per day (max 3 days)
- Attendance required for 80% of live sessions
- All work must be original; plagiarism will result in course failure

This course is designed to foster a rich learning environment and equip you for success in the field of AI and Machine Learning.

RESOURCES

Self Learning Resources

Coursera Machine Learning Specialization: https://www.coursera.org/specializations/machine-learning-introduction

GeeksforGeeks Machine Learning: https://www.geeksforgeeks.org/machine-learning/machine-learning/

Kaggle Intro to Machine Learning: https://www.kaggle.com/learn/intro-to-machine-learning

ML eBook: https://shashwatwork.github.io/assets/files/ml_ebook.pdf

Explore these resources for deeper learning and to further enhance your understanding of Artificial Intelligence and Machine Learning concepts.